

AsianNinjas

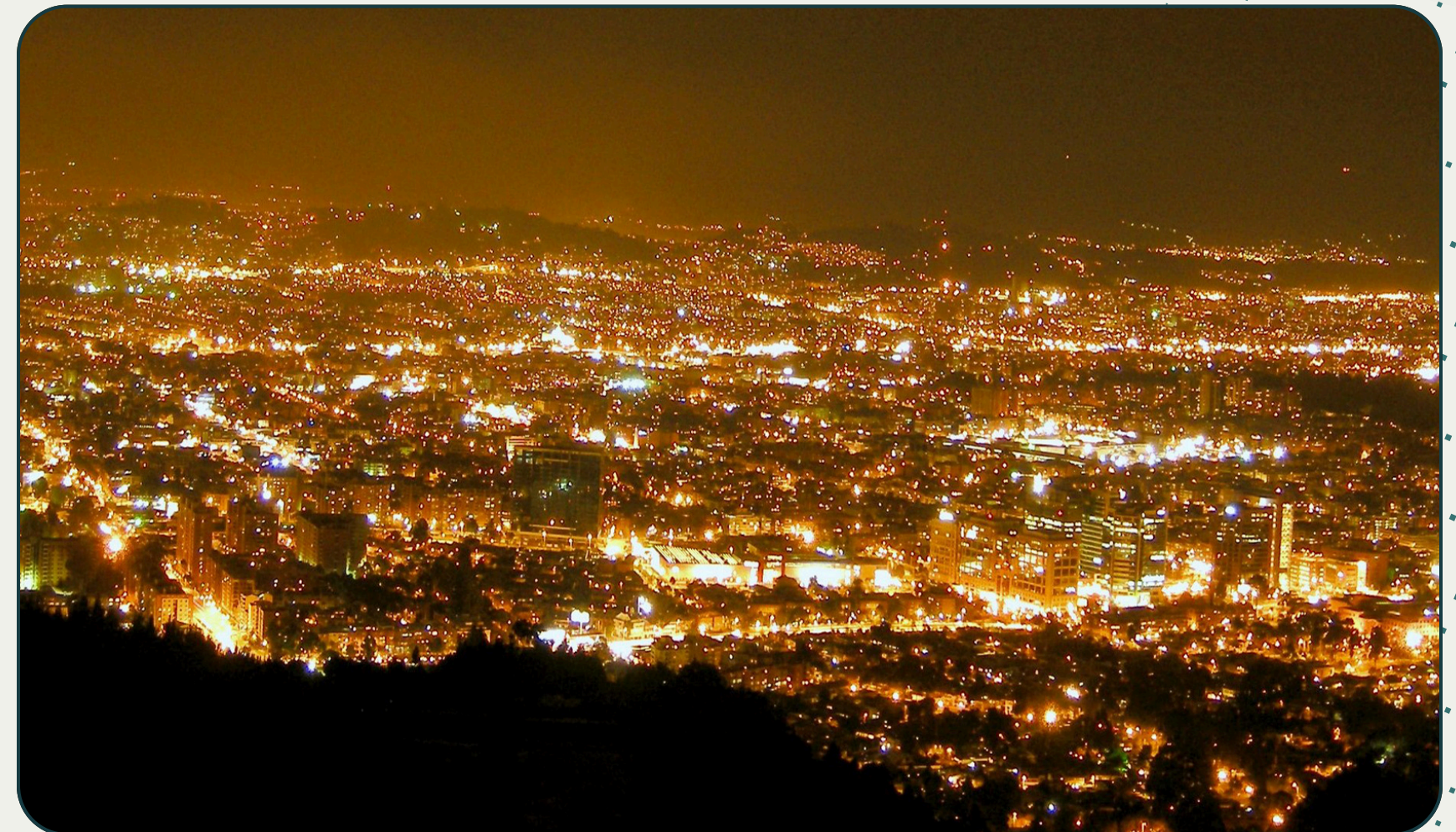
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LumiCity

LumiCity AI analyzes street-level imagery to evaluate urban lighting and infrastructure while strictly preserving citizen privacy through automated data blurring. The platform provides municipalities with an interactive map that scores street visibility, highlights blocked or missing lights, and offers AI-driven maintenance recommendations. City planners can even simulate infrastructure upgrades in real-time, ensuring public funds are spent exactly where they are needed most.



Two ways the AI runs

First, for the full city map, we process geolocated street imagery from HuggingFace, filter it to Istanbul, and run every frame through the same pipeline: anonymize, detect, score. Those results are grouped into street segments and stored in our database. That's what powers the colored map, the analytics, and the priority list.

Second, for live checks, a planner can analyze any specific location. The system fetches a Street View image, runs the same detection and scoring pipeline, and returns an instant assessment for that spot.

How AI powers LumiCity

At the heart of LumiCity is computer vision. We use a Mask2Former model trained on street-level imagery to detect what matters for lighting — lamp posts, poles, trees, buildings, sidewalks, and road geometry. The model never identifies people or vehicles. Faces and license plates are blurred before any analysis, in line with KVKK.

Tech Stack

We met every mandatory requirement: Next.js (Vercel) for the planner dashboard, Expo for the field app, and a Go backend on the required masterfabric-go clean/hexagonal architecture (deployed on Render.com), with Hugging Face models/datasets and the Google Street View API powering real AI analysis. Development stayed on-spec via Cursor rulesets (architecture, KVKK privacy, scoring consistency) so the stack and constraints were enforced across the monorepo.